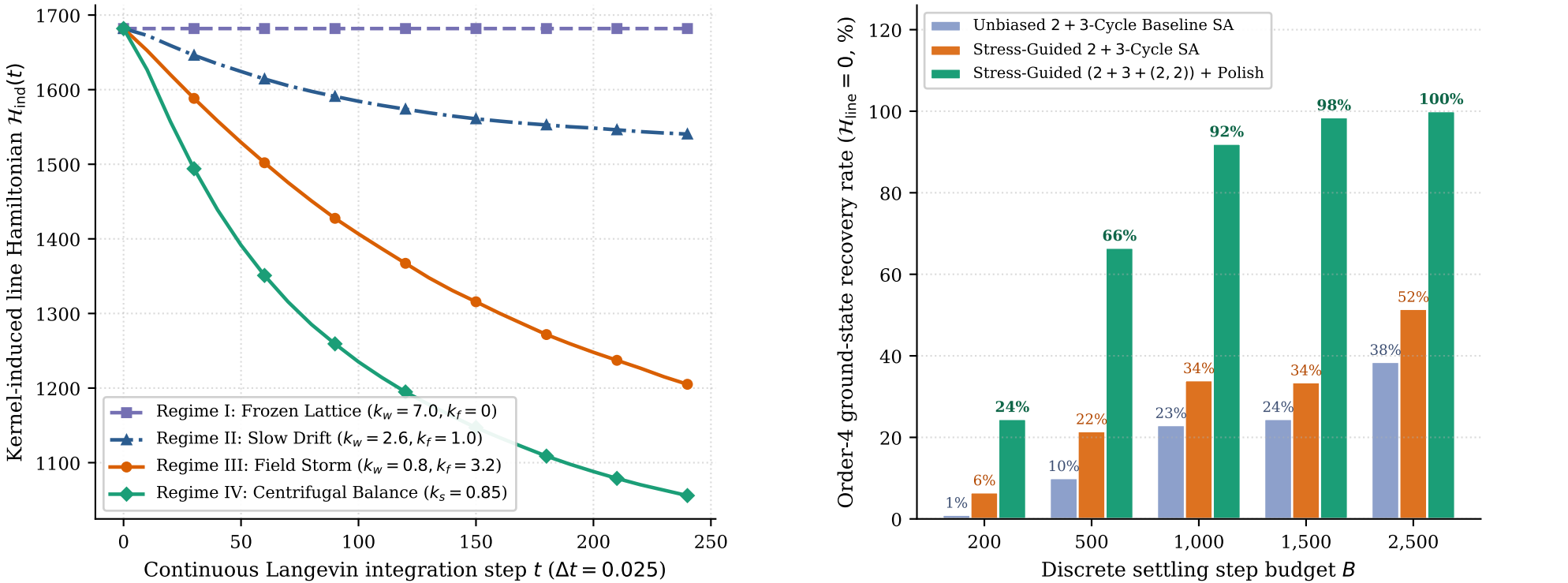
(a) Continuous Bubble-Trinity Relaxation (Regimes I--IV)
Kernel-induced line Hamiltonian $\mathcal{H}_{\text{ind}}(t)$
Continuous Langevin integration step t ($\Delta t = 0.025$)
Regime I: Frozen Lattice ($k_w = 7.0, k_f = 0$)
Regime II: Slow Drift ($k_w = 2.6, k_f = 1.0$)
Regime III: Field Storm ($k_w = 0.8, k_f = 3.2$)
Regime IV: Centrifugal Balance ($k_s = 0.85$)
(b) Order-4 Synthesis: Baseline vs. $(2, 2)$-Polished Settling
Order-4 ground-state recovery rate ($\mathcal{H}_{\text{line}} = 0$, %)
Discrete settling step budget B
Unbiased 2 + 3-Cycle Baseline SA
Stress-Guided 2 + 3-Cycle SA
Stress-Guided (2 + 3 + (2, 2)) + Polish
1%
6%
24%
10%
22%
66%
23%
34%
92%
24%
34%
98%
38%
52%
100%